

8 Robust Localization in Reverberant Rooms

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Abstract. Talker localization with microphone arrays has received significant attention lately as a means for the automated tracking of individuals in an enclosure and as a necessary component of any general purpose speech capture system. Several algorithmic approaches are available for speech source localization with multi-channel data. This chapter summarizes the current field and comments on the general merits and shortcomings of each genre. A new localization method is then presented in detail. By utilizing key features of existing methods, this new algorithm is shown to be significantly more robust to acoustical conditions, particularly reverberation effects, than the traditional localization techniques in use today.

8.1 Introduction

The primary goal of a speech localization system is accuracy. In general, estimate precision is dependent upon a number of factors. Major issues include (1) the quantity and quality of microphones employed, (2) microphone placement relative to each other and the speech sources to be analyzed, (3) the ambient noise and reverberation levels, and (4) the number of active sources and their spectral content. The performance of localization techniques generally improves with the number of microphones in the array, particularly when adverse acoustic effects are present. This has spawned the research and construction of large array systems (e.g. 512 elements) [1]. However, when acoustic conditions are favorable and the microphones are positioned judiciously, source localization can be performed adequately using a modest number (e.g. 4 elements) of microphones. Performance is clearly affected by the array geometry. The optimal design of the array based on localization criteria is typically dependent on the room layout, speaking scenarios, and the acoustic conditions [2]. In practice, many of these design considerations are very dependent on the specific application conditions, the hardware available, and non-scientific cost criteria. In an effort to make its applicability as general as possible, this chapter will focus primarily on speech localization effectiveness as a function of the acoustic degradations present, namely background noise and reverberations, rather than attempt to address more specific environmental scenarios.

In addition to high accuracy, these location estimates must be updated frequently in order to be useful in practical tracking and beamforming appli-

cations. Consider the problem of beamforming to a moving speech source. It has been shown that for sources in close proximity to the microphones, the array aiming location must be accurate to within a few centimeters to prevent high-frequency rolloff in the received signal [3] and to allow for effective channel equalization [4]. A practical beamformer must therefore be capable of including a continuous and accurate location procedure within its algorithm. This requirement necessitates the use of a location estimator capable of fine resolution at a high update rate. Additionally, any such estimator would have to be computationally non-demanding and possess a short processing latency to make it practical for real-time systems.

These factors place tight constraints on the microphone data requirements. While the computation time required by the algorithm largely determines the latency of the locator, it is the data requirements that define theoretical limits. The work in [5], for example, focuses on reducing the size of the data segments necessary for accurate source localization in realistic room environments.

The goal of this chapter is to detail the issues associated with the problem of speech source localization in reverberant and noisy rooms and to present an effective methodology for its solution. While the focus will be the single-source scenario, the techniques described, in many cases, are applicable to situations where several individuals are conversing. The more general problem of simultaneous, multi-talker localization is addressed further in Chapter 9. The following section contains a summary of the existing genres for speech source localization using microphone arrays and highlights their relative merits. It is followed in Section 8.3 by the development of a speech source localization algorithm designed specifically for reverberant enclosures which combines two of these general approaches. Section 8.4 then offers some experimental results and conclusions.

8.2 Source Localization Strategies

Existing source localization procedures may be loosely divided into three general categories: those based upon maximizing the steered response power (SRP) of a beamformer, techniques adopting high-resolution spectral estimation concepts, and approaches employing time-difference of arrival (TDOA) information. These broad classifications are delineated by their application environment and method of estimation. The first refers to any situation where the location estimate is derived directly from a filtered, weighted, and summed version of the signal data received at the sensors. The second will be used to term any localization scheme relying upon an application of the signal correlation matrix. The last category includes procedures which calculate source locations from a set of delay estimates measured across various combinations of microphones.